2.6.2 From Robustness to Adaptation

Figure 2.4 contrasts a policy trained for one robust response across the plant family with an adaptive stack that conditions its response on the current plant.

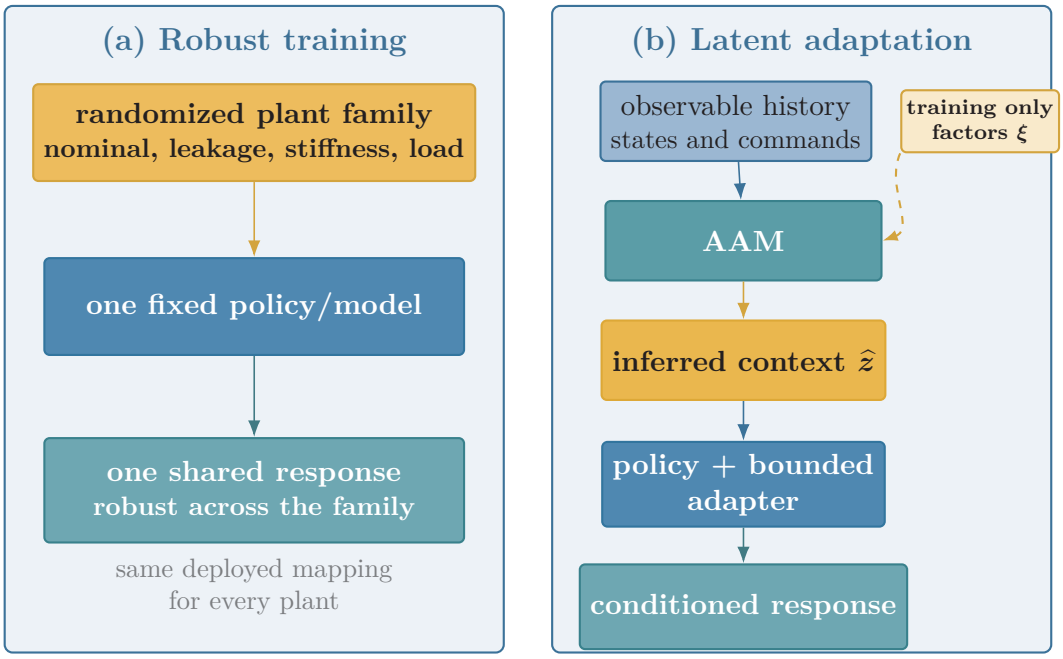


Figure 2.4: Comparison of robust training and latent adaptation. Robust training applies one fixed response across the randomized plant family. The adaptive stack uses privileged factors only to supervise training; during deployment, the AAM infers context from observable history and conditions the downstream response.

Teacher–student methods encode known simulation factors into a privileged latent and train an estimator to recover it from observable history. Lee et al. [Lee 20] used this construction for hidden terrain properties; Kumar et al. [Kumar 21] developed rapid motor adaptation. Unlike classical online parameter estimation [Åström 95], the learned latent need not have a direct physical interpretation. It can also condition a dynamics model rather than a policy, as in the contextual neural ODE of Yu et al. [Yu 26].

The AAM in Chapter 5 estimates a two-dimensional latent from one second of causal observable history. In AIF, that estimate enters the actor observation and a bounded adapter around frozen inverse dynamics; it does not condition the ROM, forward model or PhysX plant. The experiments keep two distinctions explicit: randomized training is robust but not automatically adaptive, and an informative privileged latent does not prove that an inferred latent improves closed-loop control.

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2.7 Reinforcement Learning and Structured Priors

2.7.1 Policy Optimization

Reinforcement learning optimizes a policy from observations to actions for expected cumulative reward [Sutton 18]. This thesis uses PPO [Schulman 17] with generalized